



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2016 Preliminary Course Assessment Task 2

9th of June, 2016

General instructions

- Working time – 50 minutes.
- Commence each new question on a new page.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

Teacher (please ✓)

- ☐ Mr Berry
- ☐ Mr Hwang
- ☐ Mr Ireland
- ☐ Dr Jomaa
- ☐ Miss Lee
- ☐ Mr Lin
- ☐ Ms Ziazaris

NAME # BOOKLETS USED:

Marker's use only.

QUESTION	1	2	3	4	5	Total	%
MARKS	$\overline{10}$	$\overline{10}$	$\overline{9}$	$\overline{11}$	$\overline{8}$	$\overline{48}$	

Question 1 (10 Marks)	Commence a NEW page.	Marks
(a) Express 165° in radians.		1
(b) Solve $\sin 2\theta = -1$ where $0^\circ \leq \theta \leq 360^\circ$.		3
(c) Show that $\frac{\sin x}{1 + \cos x} + \frac{1 + \cos x}{\sin x} = 2 \operatorname{cosec} x$.		3
(d) Find the exact value of $\cos \frac{7\pi}{12}$.		3

Question 2 (10 Marks)	Commence a NEW page.	Marks
(a) Find the derivative of each of the following		
i. $x^3 - 5x^2 + 3x - 1$		1
ii. $(1 - 3x^2)^5$		2
iii. $x^2\sqrt{4 - x}$		2
iv. $\frac{x + 2}{1 - 3x}$		2
(b) If $f(x) = x^2 - 2x$, use first principles to find $f'(x)$		3

Question 3 (9 Marks)	Commence a NEW page.	Marks
(a) The line m has equation $3x - 5y - 8 = 0$. The line p intersects line m at an acute angle of 45° . Find the exact gradient(s) of the line p .		3
(b) Find the coordinates of the point $P(x, y)$ which divides the interval AB externally in the ratio $1 : 3$ with $A(1, 4)$ and $B(5, 2)$.		3
(c) The equation $(x - 2y + 2) + k(x - 4y) = 0$ represents a family of straight lines passing through a fixed point P . For what value of k is the line in the family parallel to the straight line $x + y = 2$?		3

Question 4 (11 Marks)	Commence a NEW page.	Marks
(a) Find the equation of the tangent to the curve $y = 3x^2 - 10x + 6$ at the point $P(2, -2)$.		2
(b) At what point on the curve $y = (x - 2)(x + 3)$ is the tangent perpendicular to $3x - y + 5 = 0$?		3
(c) Find the shortest distance between the lines $y = -\frac{1}{4}x + 1$ and $x + 4y + 12 = 0$.		3
(d) Sketch the region $y < \sqrt{3 - x^2}$.		3

Question 5 (8 Marks)	Commence a NEW page.	Marks
(a) The angle of elevation of a hill OT , at a point A due south of it is 20° , and at point B which is on a bearing of $060^\circ T$ from O , the angle of elevation is 40° . The distance AB is 500m. Draw a diagram to represent all the given information. Find the height of the hill to the nearest metre.		4
(b) i. Show that $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$.		1
ii. Hence solve $\sin 3x + \sin 4x + \sin 5x = 0$ for $0^\circ \leq x \leq 180^\circ$.		3

End of paper.

2016 Y11 AT2 Suggested Solutions

Marks

Question 1

1 a) $165 \times \frac{\pi}{180} = \frac{11\pi}{12}$

3 b) $\sin 2\theta = -1 \quad 0 \leq \theta \leq 360$
 $0 \leq 2\theta \leq 720$

$$2\theta = 270, 630$$

$$\theta = 135, 315$$

3 c) $LHS = \frac{\sin x}{1 + \cos x} + \frac{1 + \cos x}{\sin x}$

$$= \frac{\sin^2 x + (1 + \cos x)^2}{\sin x (1 + \cos x)}$$

$$= \frac{\sin^2 x + 1 + 2\cos x + \cos^2 x}{\sin x (1 + \cos x)}$$

$$= \frac{2 + 2\cos x}{\sin x (1 + \cos x)}$$

$$= \frac{2(1 + \cos x)}{\sin x (1 + \cos x)}$$

$$= \frac{2}{\sin x}$$

$$= 2 \operatorname{cosec} x$$

$$= RHS$$

3 d) $\cos \frac{7\pi}{12} = \cos \left(\frac{\pi}{4} + \frac{\pi}{3} \right)$

$$= \cos \frac{\pi}{4} \cos \frac{\pi}{3} - \sin \frac{\pi}{4} \sin \frac{\pi}{3}$$

$$= \frac{1}{\sqrt{2}} \times \frac{1}{2} - \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2}$$

$$= \frac{1 - \sqrt{3}}{2\sqrt{2}}$$

Marks

Question 2

1 a) i) $3x^2 - 10x + 3$

2 ii) $5(1-3x^2)^4 \cdot (-6x)$
 $= -30x(1-3x^2)^4$

2 iii) $u = x^2$ $v = (4-x)^{\frac{1}{2}}$
 $u' = 2x$ $v' = \frac{1}{2}(4-x)^{-\frac{1}{2}} \cdot (-1)$

$$y' = 2x\sqrt{4-x} - \frac{x^2}{2\sqrt{4-x}}$$

2 iv) $u = x+2$ $v = 1-3x$
 $u' = 1$ $v' = -3$

$$y' = \frac{1-3x+3(x+2)}{(1-3x)^2}$$

$$= \frac{7}{(1-3x)^2}$$

3 b) $f(x) = x^2 - 2x$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 2(x+h) - x^2 + 2x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 2x - 2h - x^2 + 2x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x + h - 2)}{h}$$

$$= \lim_{h \rightarrow 0} (2x + h - 2)$$

$$= 2x - 2$$

Marks

3 a)

Question 3

$$3x - 5y - 8 = 0$$

$$m_2 = \frac{3}{5}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan 45 = \left| \frac{m - \frac{3}{5}}{1 + \frac{3m}{5}} \right|$$

$$\frac{m - \frac{3}{5}}{1 + \frac{3m}{5}} = 1$$

$$\frac{m - \frac{3}{5}}{1 + \frac{3m}{5}} = -1$$

$$\frac{5m - 3}{5 + 3m} = 1$$

$$\frac{5m - 3}{5 + 3m} = -1$$

$$5m - 3 = 5 + 3m$$

$$5m - 3 = -5 - 3m$$

$$2m = 8$$

$$8m = -2$$

$$m = 4$$

$$m = -\frac{1}{4}$$

3 b)

$$-1 : 3$$

$$A(1, 4) \quad B(5, 2)$$

$$P(x, y) = \left(\frac{-5 + 3}{-1 + 3}, \frac{-2 + 12}{-1 + 3} \right)$$

$$= (-1, 5)$$

Marks

3 c) $(x - 2y + 2) + k(x - 4y) = 0$

$$m = - \frac{(-2 - 4k)}{(1 + k)}$$

$$-1 = \frac{2 + 4k}{1 + k}$$

$$-1 - k = 2 + 4k$$

$$-3 = 5k$$

$$k = -\frac{3}{5}$$

Marks

2

a)

Question 4

$$y' = 6x - 10$$

$$m = 12 - 10$$

$$= 2$$

Eq'n of tangent:

$$y + 2 = 2(x - 2)$$

$$y = 2x - 6$$

3

b)

$$3x - y + 5 = 0$$

$$m = 3$$

$$m_{\perp} = -\frac{1}{3}$$

$$y = (x - 2)(x + 3)$$

$$= x^2 + x - 6$$

$$y' = 2x + 1$$

$$2x + 1 = -\frac{1}{3}$$

$$6x + 3 = -1$$

$$6x = -4$$

$$x = -\frac{2}{3}$$

$$y = \left(-\frac{2}{3}\right)^2 + \left(-\frac{2}{3}\right) - 6$$

$$= -\frac{56}{9}$$

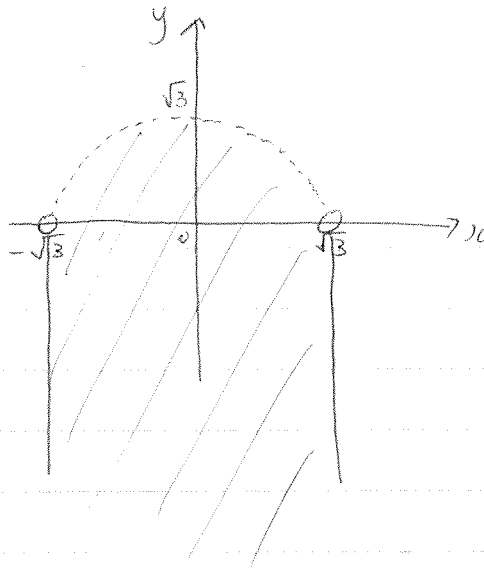
$\therefore$ point is $\left(-\frac{2}{3}, -\frac{56}{9}\right)$

Marks

3 c) $(0, 1)$ lies on $y = -\frac{1}{4}x + 1$

$$d_{\perp} = \frac{|1(0) + 4(1) + 12|}{\sqrt{1^2 + 4^2}}$$
$$= \frac{16}{\sqrt{17}}$$

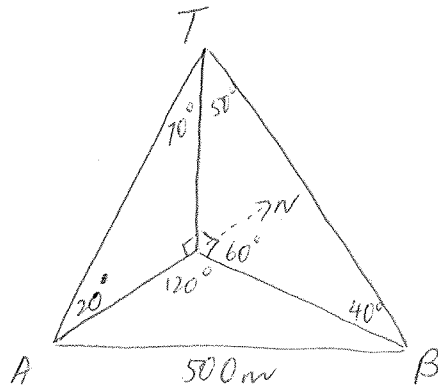
3 d)



Marks

4 a)

Question 5



$$\tan 70^\circ = \frac{AO}{h}$$

$$AO = h \tan 70$$

$$\tan 50^\circ = \frac{BO}{h}$$

$$BO = h \tan 50$$

$$\begin{aligned} 500^2 &= (h \tan 70)^2 + (h \tan 50)^2 - 2(h \tan 70)(h \tan 50) \cos 120 \\ &= h^2 [\tan^2 70 + \tan^2 50 - 2 \tan 70 \tan 50 \cos 120] \end{aligned}$$

$$h > 0$$

$$\begin{aligned} h &= \sqrt{\frac{500^2}{\tan^2 70 + \tan^2 50 - 2 \tan 70 \tan 50 \cos 120}} \\ &= 143 \text{ m} \end{aligned}$$

1 b) i) $LHS = \sin(A+B) + \sin(A-B)$

$$= \sin A \cos B + \cos A \sin B + \sin A \cos B - \cos A \sin B$$

$$= 2 \sin A \cos B$$

$$= RHS$$

3 ii) $\sin 3x + \sin 4x + \sin 5x = \sin(4x-x) + \sin 4x + \sin(4x+x)$

$$\sin(4x-x) + \sin(4x+x) = 2 \sin 4x \cos x$$

$$2 \sin 4x \cos x + \sin 4x = 0$$

$$0 \leq x \leq 180$$

$$\sin 4x (2 \cos x + 1) = 0$$

$$0 \leq 4x \leq 720$$

$$\sin 4x = 0$$

$$2 \cos x + 1 = 0$$

$$4x = 0, 180, 360, 540, 720$$

$$\cos x = -\frac{1}{2}$$

$$x = 0, 45, 90, 135, 180$$

$$x = 120$$

$$\therefore x = 0^\circ, 45^\circ, 90^\circ, 120^\circ, 135^\circ, 180^\circ$$